



WRITTEN EXAM

Course Name: Modellering och reglering / Modeling and Control	Course Code: R0004E
Number of examination hours: 5	Date: 2024-10-24

Allowed aids:

- non-programmed calculator
- "formelsamling i reglerteknik" document
- an A4-sheet (both sides) with own notes that are submitted with the solution sheets

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Grade scale: 3 – 12 points, 4 – 16 points, 5 – 20 points	Number of questions and total score: 7 questions, 25 points
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Other information:

General instructions

Check that you have received all the exams/questions.

All new answers begin on a separate page. Do not write on the back page.

Use the provided semi-log paper to draw the Bode plot.

Print, write clearly, and do not use red pen.

After examination

Examination results will be announced within three weeks from the examination date and no later than two weeks before next re-examination.

The exam result can be seen at My LTU – Ladok for students.

Your corrected exam is scanned and visible on My LTU – Graded exams.

Information to the printer

Project number: 341980	Number of copies 52	Number of pages 4
Other information: Please attach the Appendices on each copy		

-
1. (a) Mention and explain the two main approaches to develop a model of a dynamical system. (0.5)
 - (b) Based on observation of the step response, most dynamical systems can be approximated as either a first order or a second order system. Specify the main difference between the first and second order system, based on its response. Please explain using graphical illustration to help. (1.5)
 - (c) Write the standard transfer function of first order and second order systems and list the main parameters of each transfer function. (1)

Total for Question 1: 3 points

2. A shaker table system is modeled using mass-spring-damper elements, as shown in Figure 1. The mass of the table is M , the mass of the camera is m , the spring stiffness constant is K_s , the damping constant is B . The measured quantity is the displacement of the table and the input to the system is the force $f_i(t)$. Using Newton's second law, the dynamics of the system can be represented by the second-order differential equation

$$a\ddot{x}(t) + b\dot{x}(t) + cx(t) = u(t)$$

where $x(t)$ is the displacement and $u(t)$ is the input force from the coil.

- (a) Substitute the parameters a , b and c with the physical parameters of the system and determine the input and the output of the system. (1)
- (b) Derive the state-space model of the system, by finding the matrices A , B , C and D , and write the state-space representation of the system. (2)

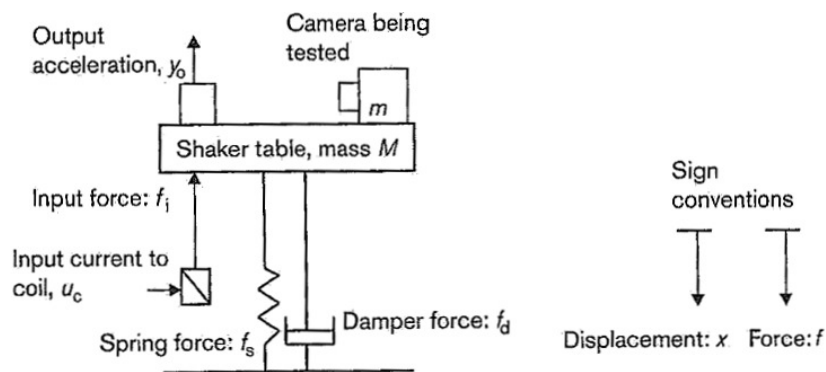


Figure 1: Schematic diagram of a shaker table and forces acting on it

- (c) Given $m = 250$ g, $K_s = 1.1 \times 10^7$ N/m and $B = 1100$ N/ms, determine the range of values of M such that the system is stable. Provide explanation to your answer. (1)

Total for Question 2: 4 points

3. Given the following transfer function of a plant:

$$G(s) = \frac{10}{(s+2)(s+3)^2}$$

- (a) Find the Laplace inverse of the output function $Y(s)$, if the plant $G(s)$ is excited by an input $R(s)$ which is an impulse. (2)
- (b) Find the steady-state value (final value) of the output $y(t)$. (1)

Total for Question 3: 3 points

4. Consider the the following transfer function of a plant:

$$G(s) = \frac{6}{s^2 + 4.2s + 36}$$

- (a) Find the closed-loop transfer function $T(s) = \frac{Y(s)}{R(s)}$, if the feedback is a unity feedback. (1)
- (b) Find the steady state error of each the open-loop system $G(s)$ and the closed-loop system $T(s)$, with respect to a unit step input ($R(s) = \frac{1}{s}$). (1)
- (c) Explain your answer by relating the steady-state errors that you find, to the type of the system. (1)
- (d) Compare the steady state errors that you obtained in (b), which one is larger and give explanation. (1)

Total for Question 4: 4 points

5. The open-loop transfer function of a system is given by

$$G(s) = \frac{K(s^2 - 8s + 15)}{s^2 + 2s + 8}$$

For $K = 1$, the Nyquist plot of $G(s)$ is shown in Figure 2.

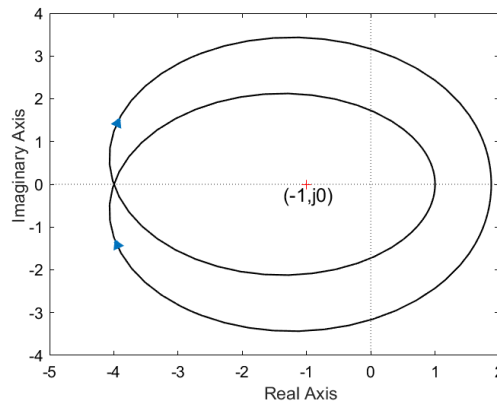


Figure 2: Nyquist plot for $G(s)$ with $K = 1$.

- (a) Considering positive frequency, determine where the Nyquist plot starts (i.e. the value of $|G(j\omega)|$ and $\angle G(j\omega)$ when $\omega = 0$) and where the plot ends (i.e. the value of $|G(j\omega)|$ and $\angle G(j\omega)$ when $\omega = \infty$). (1.5)
- (b) Find the value of N , P and Z , and determine the stability of the closed-loop system using the Nyquist criterion. (1.5)

Total for Question 5: 3 points

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6. Consider the position control system in Figure 3, in which K is an adjustable gain parameter and

$$G(s) = \frac{(s + 3)}{(s + 2)(s^2 + 2s + 25)}$$

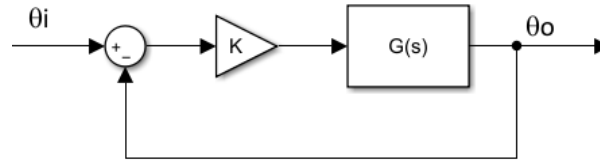


Figure 3: Block diagram of a position control system.

- On the semilog graph paper(s) provided, sketch the Bode plot, both the magnitude and phase plots, of the open-loop transfer function $KG(s)$, with $K = 2$. Include the sketch of the Bode plot of each term. (3)
- Find the gain margin (GM) and the phase margin (PM) of the Bode plot. (0.5)
- Explain whether the system is stable or not, based on the value of GM and PM that you obtain in (b). (0.5)

Total for Question 6: 4 points

7. The open-loop transfer function of a 2nd order plant is given by

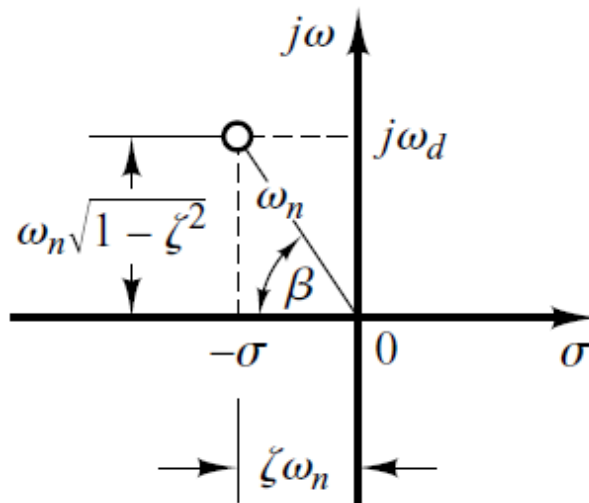
$$G(s) = \frac{6}{s^2 + 2s + 6}$$

- A PD controller ($K_p + K_d s$) is added in series with the plant. Find the closed-loop transfer function of the system and write the closed-loop characteristic equation in terms of K_p and K_d . (1)
- Find the 2nd order system parameters in terms of K_p and K_d , for the closed-loop transfer function. (0.5)
- Determine the value of the proportional gain K_p and the differential gain K_d , such that the 2nd order parameters of the closed loop system has the values $\omega_n = 10$ and $\zeta = 0.9$. (2)
- Explain what the expected effect of the use of the PD controller in this system. (0.5)

Total for Question 7: 4 points

END

1. $t_r = \frac{\pi - \beta}{\omega_d}$
2. $t_p = \frac{\pi}{\omega_d}$
3. $M_p = e^{-\frac{\zeta \pi}{\sqrt{1 - \zeta^2}}}$
4. $t_s = \frac{4}{\zeta \omega_n}$ (2% criteria) or $t_s = \frac{3}{\zeta \omega_n}$ (5% criteria)
5. Definition of the angle β :



6. Variation of unit step response of a second order system with respect to the value of ζ :

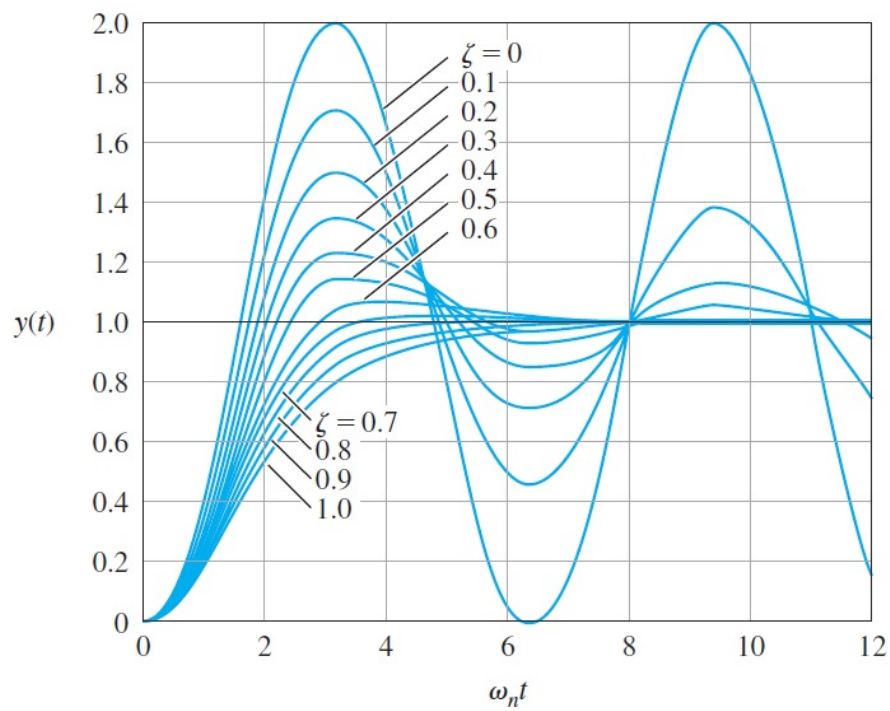
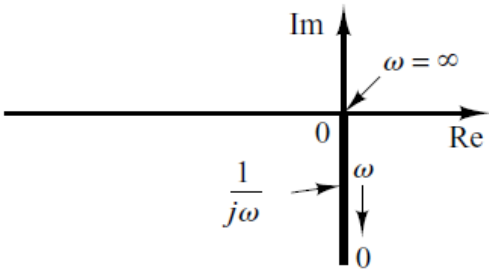
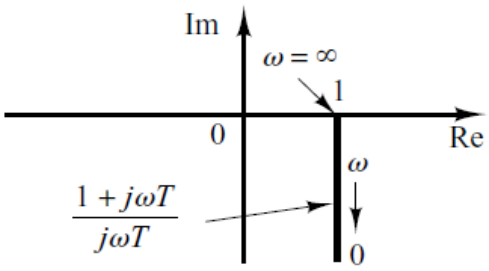
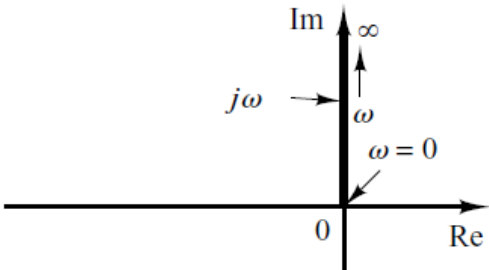
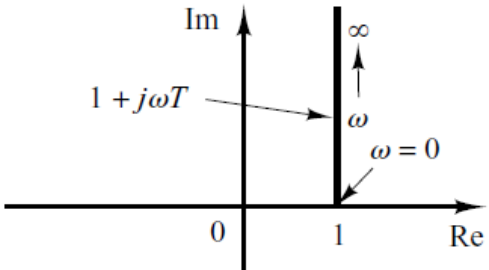
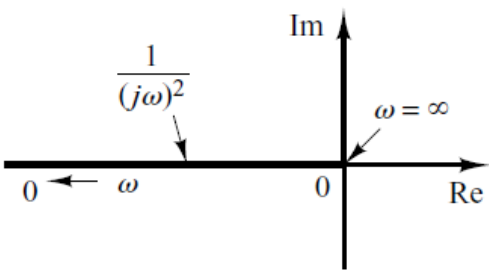
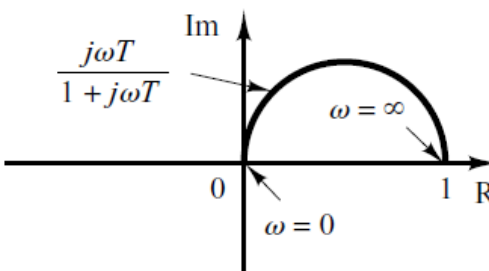
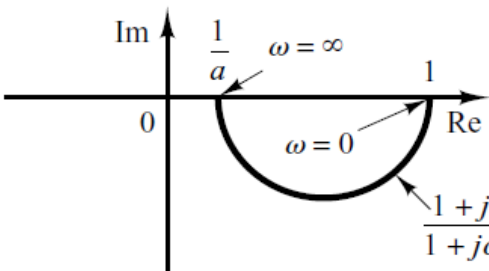
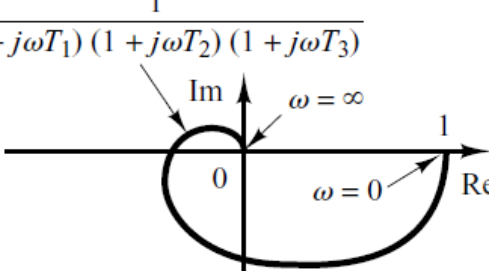
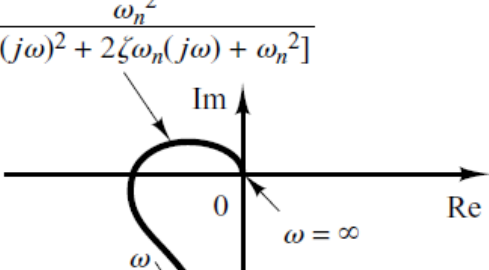
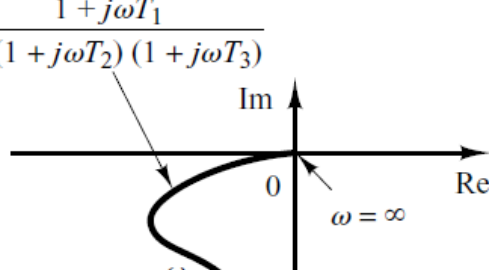


Table of Laplace Transforms

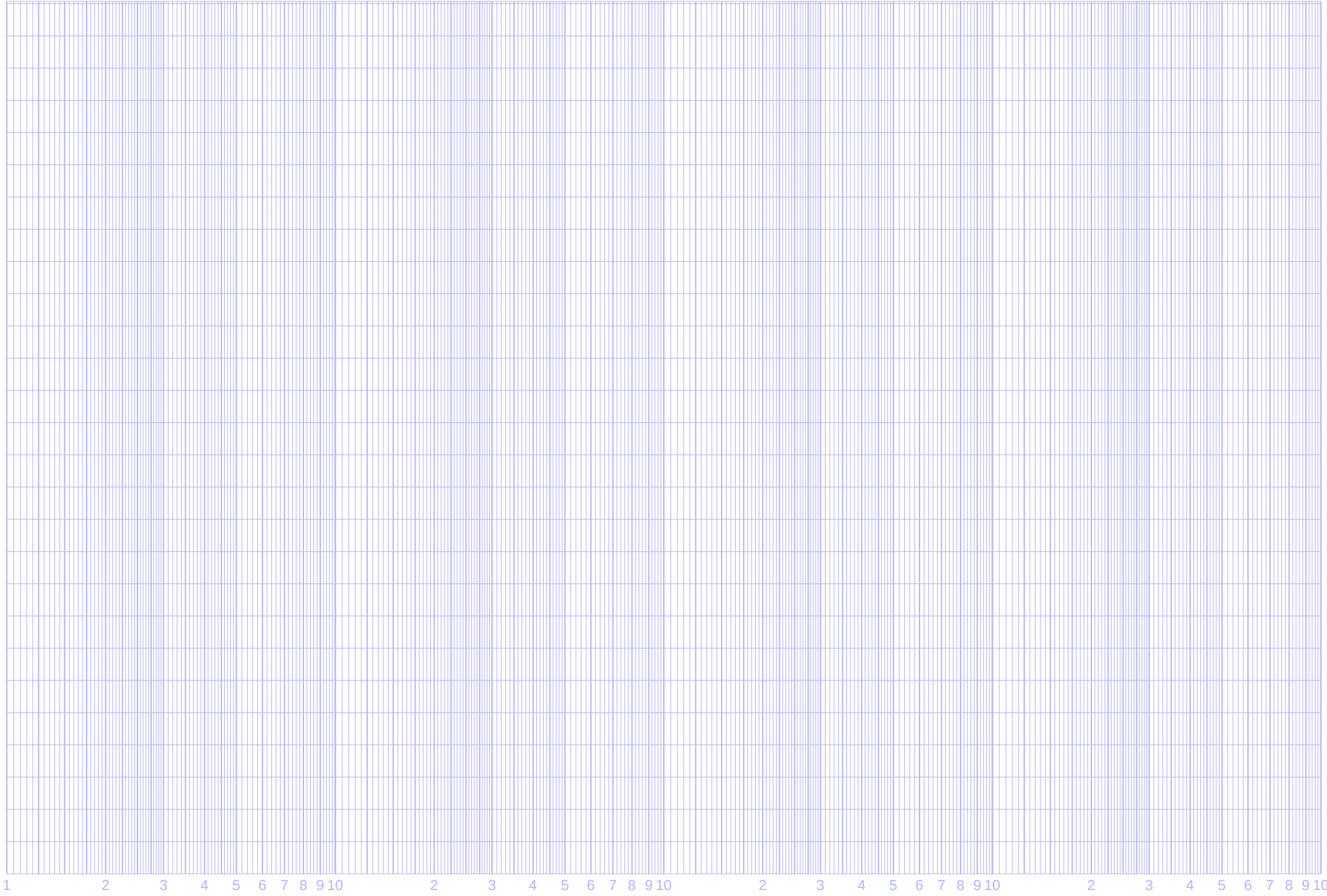
$f(t) = \mathcal{L}^{-1}\{F(s)\}$	$F(s) = \mathcal{L}\{f(t)\}$	$f(t) = \mathcal{L}^{-1}\{F(s)\}$	$F(s) = \mathcal{L}\{f(t)\}$
1. 1	$\frac{1}{s}$	2. e^{at}	$\frac{1}{s-a}$
3. $t^n, \quad n=1,2,3,\dots$	$\frac{n!}{s^{n+1}}$	4. $t^p, p > -1$	$\frac{\Gamma(p+1)}{s^{p+1}}$
5. $\sqrt{t}$	$\frac{\sqrt{\pi}}{2s^{\frac{3}{2}}}$	6. $t^{n-\frac{1}{2}}, \quad n=1,2,3,\dots$	$\frac{1 \cdot 3 \cdot 5 \cdots (2n-1)\sqrt{\pi}}{2^n s^{n+\frac{1}{2}}}$
7. $\sin(at)$	$\frac{a}{s^2+a^2}$	8. $\cos(at)$	$\frac{s}{s^2+a^2}$
9. $t \sin(at)$	$\frac{2as}{(s^2+a^2)^2}$	10. $t \cos(at)$	$\frac{s^2-a^2}{(s^2+a^2)^2}$
11. $\sin(at) - at \cos(at)$	$\frac{2a^3}{(s^2+a^2)^2}$	12. $\sin(at) + at \cos(at)$	$\frac{2as^2}{(s^2+a^2)^2}$
13. $\cos(at) - at \sin(at)$	$\frac{s(s^2-a^2)}{(s^2+a^2)^2}$	14. $\cos(at) + at \sin(at)$	$\frac{s(s^2+3a^2)}{(s^2+a^2)^2}$
15. $\sin(at+b)$	$\frac{s \sin(b) + a \cos(b)}{s^2+a^2}$	16. $\cos(at+b)$	$\frac{s \cos(b) - a \sin(b)}{s^2+a^2}$
17. $\sinh(at)$	$\frac{a}{s^2-a^2}$	18. $\cosh(at)$	$\frac{s}{s^2-a^2}$
19. $e^{at} \sin(bt)$	$\frac{b}{(s-a)^2+b^2}$	20. $e^{at} \cos(bt)$	$\frac{s-a}{(s-a)^2+b^2}$
21. $e^{at} \sinh(bt)$	$\frac{b}{(s-a)^2-b^2}$	22. $e^{at} \cosh(bt)$	$\frac{s-a}{(s-a)^2-b^2}$
23. $t^n e^{at}, \quad n=1,2,3,\dots$	$\frac{n!}{(s-a)^{n+1}}$	24. $f(ct)$	$\frac{1}{c} F\left(\frac{s}{c}\right)$
25. $u_c(t) = u(t-c)$ Heaviside Function	$\frac{e^{-cs}}{s}$	26. $\delta(t-c)$ Dirac Delta Function	e^{-cs}
27. $u_c(t) f(t-c)$	$e^{-cs} F(s)$	28. $u_c(t) g(t)$	$e^{-cs} \mathcal{L}\{g(t+c)\}$
29. $e^{ct} f(t)$	$F(s-c)$	30. $t^n f(t), \quad n=1,2,3,\dots$	$(-1)^n F^{(n)}(s)$
31. $\frac{1}{t} f(t)$	$\int_s^\infty F(u) du$	32. $\int_0^t f(v) dv$	$\frac{F(s)}{s}$
33. $\int_0^t f(t-\tau) g(\tau) d\tau$	$F(s)G(s)$	34. $f(t+T) = f(t)$	$\frac{\int_0^T e^{-st} f(t) dt}{1-e^{-sT}}$
35. $f'(t)$	$sF(s) - f(0)$	36. $f''(t)$	$s^2 F(s) - sf(0) - f'(0)$
37. $f^{(n)}(t)$	$s^n F(s) - s^{n-1} f(0) - s^{n-2} f'(0) \cdots - sf^{(n-2)}(0) - f^{(n-1)}(0)$		

Table 7-1 Polar Plots of Simple Transfer Functions

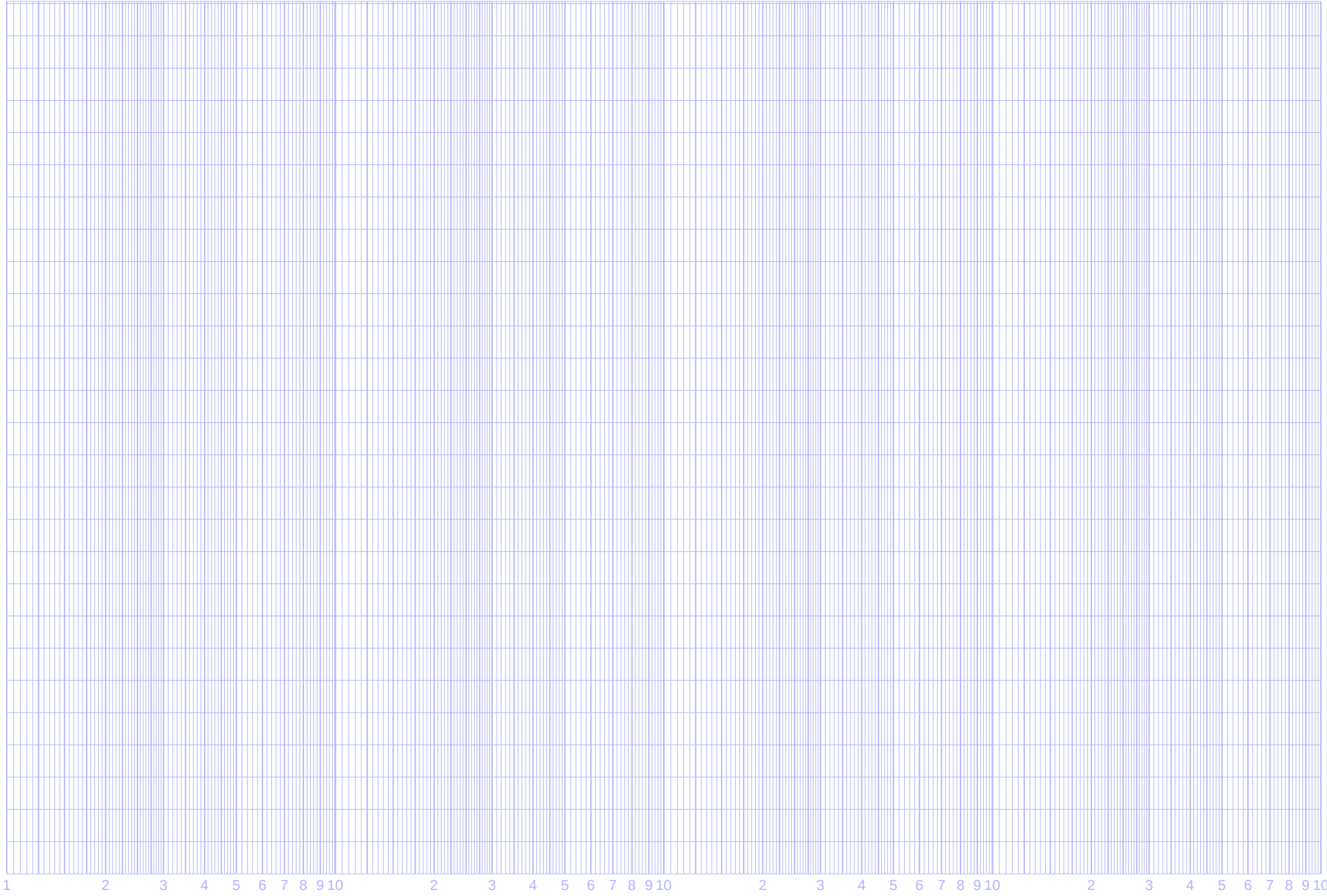
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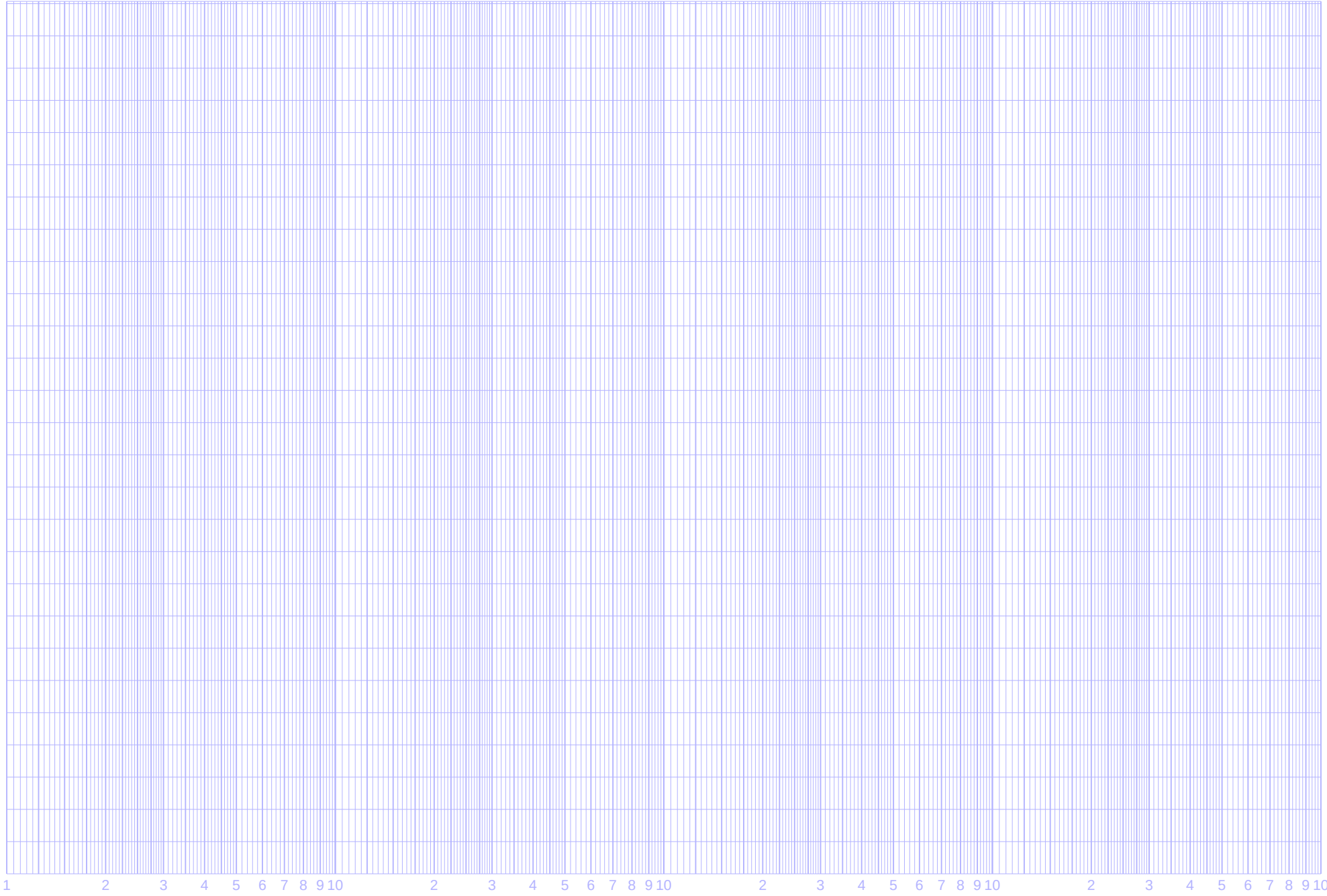
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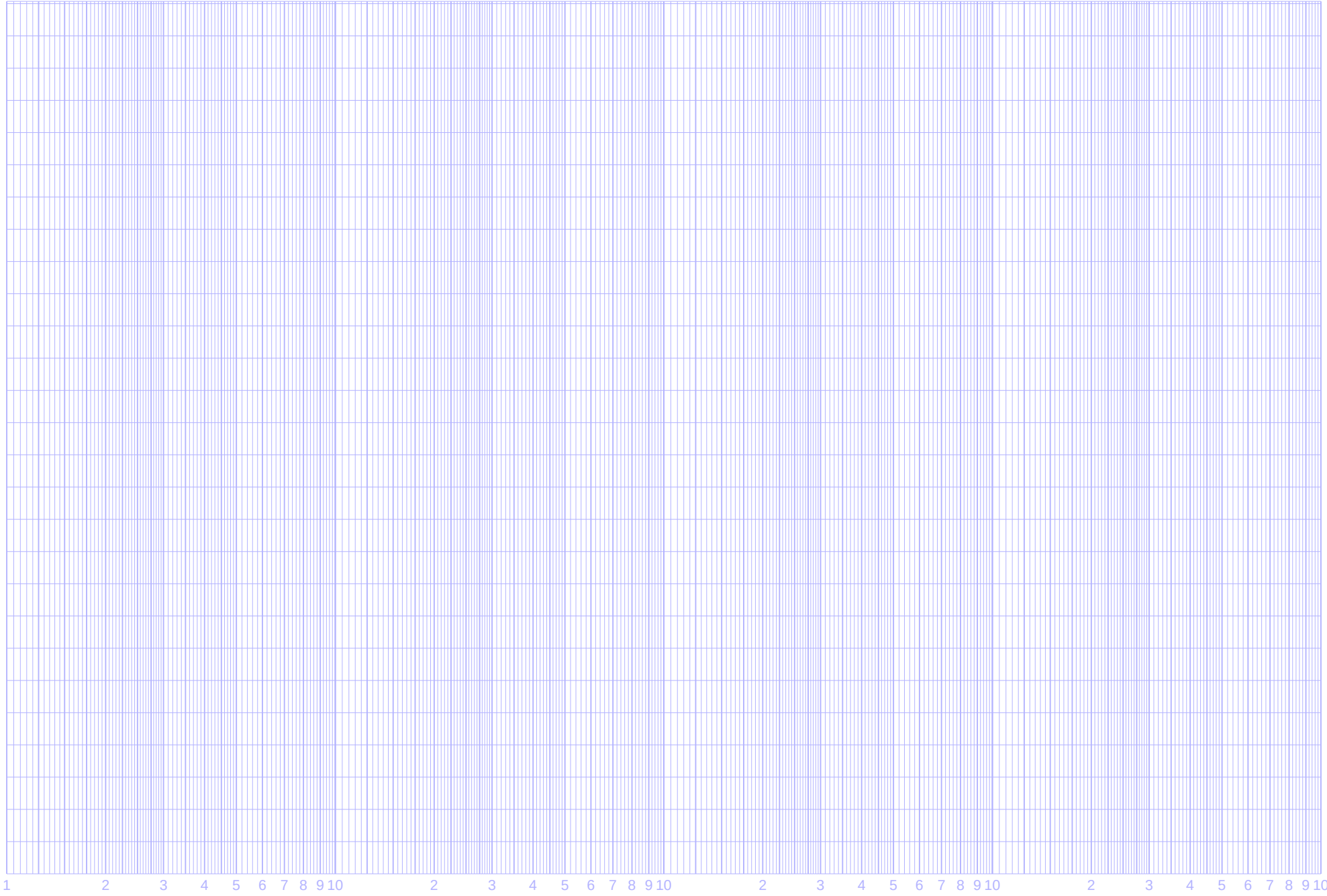
PERSONNUMMER:

R0003E / R0004E



PERSONNUMMER:

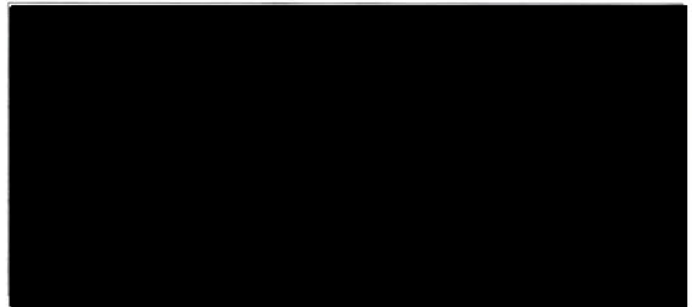
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KURSKOD / COURSE CODE	PROV / TEST CODE
R 0 0 0 4 E	0 0 0 3
KURSBENÄMNING / COURSE NAME	
Modellering och reglering	
PROVBENÄMNING / TEST NAME	
Tentamen	
TENTAMENSdatum / EXAMINATION DATE	
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TENTAMENSORT/CITY (för distansstudenter / for off campus students only)	

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PROGRAM	INSKR.ÅR/YEAR	ANTAL SIDOR / NO. OF PAGES
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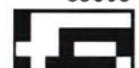
Skanningsblad/Scanning Sheet

Behandlat
uppgift nr (sätt x) /
Mark the questions you
answered with an X

Lärarens anteckningar / Teacher's notes

1	X	2	
2	X	4	
3	X	1.5	
4	X	3.5	
5	X	3	
6	X	3.3	
7	X	1.8	
8			
9			
10			
11			
12			
13			
14			
15			
16			
17			
18			
Poängsumma Points		19.1	Betyg Grade 4

35065



a) Well a dynamical system
is a system that has
elements or components
that change with time
2 approaches are Pd and Pi
systems.

b) 2 order system oscillate
while 1st order doesn't ✓
really so that

c)
$$\frac{W_n}{s^2 + 2\zeta W_n s + W_n^2} \quad (2nd)$$

Main params are W_n and ζ

W_n = Natural freq

ζ = damping factor

first order?

Uppgift nr:

1

Poäng:

2

Lärarens
anteckning:

✓

1.5

0.5

a)

$$a\ddot{x} + b\dot{x} + cx = u$$

$$x = m, \dot{x} = m/s, \ddot{x} = m/s^2$$

We know that

$$F_{res} = ma \quad a = m/s^2 \text{ (Newton II Law)}$$

$$F_s = kx \quad x = m \text{ (Spring force)}$$

$$F_d = B\dot{x} \quad \dot{x} = m/s \text{ (Damping force)}$$

The total mass will be
(m + M)

With this information we can
substitute in the variables.

$$(m+M)\ddot{x} + B\dot{x} + k_s x = f_i(t) \quad \checkmark$$

$$u(t) = f_i(t) \text{ (input force)}$$

b)

New variables

$$\dot{x}_1 = x_2, \quad \dot{x}_2 = \ddot{x}, \quad x_1 = x, \quad x_2 = \dot{x}$$

$$(m+M)\dot{x}_2 + Bx_2 + k_s x_1 = f_i(t)$$

$$\dot{x}_2 = \frac{f_i(t) - Bx_2 - k_s x_1}{(m+M)}$$

$$\dot{x} = \begin{bmatrix} 0 & 1 \\ -\frac{k_s}{(m+M)} & -\frac{B}{(m+M)} \end{bmatrix} x + \begin{bmatrix} 0 \\ 1 \\ (m+M) \end{bmatrix} u \quad f_i(t) = u \quad \checkmark$$

$$y = [C_1 \ C_2] x + D u \quad D = 0 \quad \checkmark$$

we can set $C_1 = 1$ and $C_2 = 0$

Uppgift nr:

2.

Poäng:

3

Lärarens
anteckning:

1

2

9

$$m = 250 \text{ g} \quad k_s = 1.1 \cdot 10^7 \text{ N/m} \quad B = 1100 \text{ N/m/s}$$

System is stable when

$$\det(SI - A) = 0$$

When the real parts of the poles are negative.

$$SI = \begin{bmatrix} s & 0 \\ 0 & s \end{bmatrix} \quad A = \begin{bmatrix} 0 & 1 \\ -\frac{k_s}{(m+M)} & -\frac{B}{(m+M)} \end{bmatrix}$$

$$SI - A = \begin{bmatrix} s & -1 \\ \frac{k_s}{(m+M)} & s + \frac{B}{(m+M)} \end{bmatrix}$$

$$\det(SI - A) = s \left(s + \frac{B}{(m+M)} \right) + \frac{k_s}{(m+M)}$$

$$s^2 + \frac{sB}{(m+M)} + \frac{k_s}{(m+M)} = 0$$

$$p, q: s = -\frac{B}{2 \cdot (m+M)} \pm \sqrt{\left(\frac{B}{2 \cdot (m+M)}\right)^2 - \frac{k_s}{(m+M)}}$$

$$= -\frac{550}{(0.25+M)} \pm \sqrt{\left(\frac{550}{(0.25+M)}\right)^2 - \frac{1.1 \cdot 10^7}{(0.25+M)}}$$

The square root part will always be negative. which means both real parts will be negative as long as $M > 0$.

Since we don't want the mass to be negative.

Uppgift nr:

3

Poäng:

1.5

Lärarens

anteckning:

a) $G(s) = \frac{10}{(s+2)(s+3)^2}$

$$G(s) = \frac{Y(s)}{R(s)}$$

$$\frac{Y(s)}{R(s)} = \frac{10}{(s+2)(s+3)^2} \cdot R(s)$$

$$Y(s) = \frac{10}{(s+2)(s+3)^2}$$

$$\frac{10}{(s+2)(s+3)^2} \stackrel{L^{-1}}{=} \frac{1}{s(s^2+2\omega_0 s + \omega_0^2)}$$

$$\frac{1}{10} \cdot \frac{1}{s(s^2+6s^2+21)-18}$$

$$2\omega_0 = 6 \quad \omega_0^2 = 21$$

$$\gamma = \frac{6}{2\sqrt{21}} \approx 0.659$$

Use Partial
fraction
expansion!

$$0 < \gamma < 1: 1 - \frac{1}{\sqrt{1-\gamma^2}} e^{-\gamma\omega_0 t} \sin(\omega_0 \sqrt{1-\gamma^2} t + \phi)$$

$$y(t) = 1 - 1.32 e^{-3t} \sin(3.966 \dots t + \phi)$$

$$\phi = \arccos \gamma$$

b) steady state value

$$\lim_{t \rightarrow \infty} y(t) = \lim_{t \rightarrow \infty} e^{-3 \cdot \infty} \approx 0$$

1

Uppgift nr:

4

Poäng:

2

Lärarens
anteckning:

$$a) \quad T(s) = \frac{G(s)}{1 + G(s)}$$

$$G(s) = \frac{6}{s^2 + 4.2s + 36}$$

$$T(s) = \frac{6}{s^2 + 4.2s + 36} \cdot \frac{1}{1 + \frac{6}{s^2 + 4.2s + 36}}$$

$$= \frac{6}{s^2 + 4.2s + 36} \cdot \frac{s^2 + 4.2s + 36}{s^2 + 4.2s + 36 + 6}$$

$$\underline{T(s)} = \frac{6}{s^2 + 4.2s + 42} \quad \checkmark$$

b) Open loop system:

We can use FVT.

$$\lim_{s \rightarrow 0} s G(s) \cdot \frac{1}{s} = \lim_{s \rightarrow 0} \frac{6}{s^2 + 4.2s + 36} = \frac{6}{36}$$

$$E_{ss} = R(s) - Y(s) \\ = 1 - \frac{6}{36} = \frac{30}{36} = \frac{5}{6} = 0.8333.. \quad \checkmark$$

Closed loop system:

$$E_{ss} = \frac{1}{1 + K_p}$$

$$K_p = \lim_{s \rightarrow 0} T(s) = \frac{6}{s^2 + 4.2s + 42} = \frac{6}{42}$$

$$E_{ss} = \frac{1}{1 + \frac{6}{42}} = \frac{1}{\frac{48}{42}} = 0.875 \quad \checkmark$$

c)

Since the steady state error we got was a finite value ✓
I believe its a type 0 system.

d)

the Closed loop TCS steady error rate was slightly bigger.
Which is caused by the unity feedback and potentially measurement noise.

Uppgift nr:

4

Poäng:

1.5

Lärarens anteckning:

1

0.5

a)

$$G(s) = \frac{k(s^2 - 8s + 15)}{s^2 + 2s + 8}$$

$$k=1$$

$$G(j\omega) = \frac{(j\omega)^2 - 8j\omega + 15}{(j\omega)^2 + 2j\omega + 8}$$

$$\lim_{\omega \rightarrow 0} G(j\omega) = \frac{15}{8} \quad \checkmark$$

$$|G(0)| = \frac{15}{8}$$

$$\angle G(0) = 0^\circ \quad \checkmark$$

$$\lim_{\omega \rightarrow \infty} G(j\omega) = \frac{(j\omega)^2}{(j\omega)^2} = 1$$

Since $(j\omega)^2$ are the fastest growing terms.

$$\lim_{\omega \rightarrow \infty} |G(j\omega)| = 1 \quad \checkmark$$

$$\angle G(j\omega) = 0^\circ \quad \checkmark$$

b)

$$Z = N + P$$

P = Poles in RHP

N = clockwise encirclement around $(-1, j\omega)$

$$\text{Poles: } s^2 + 2s + 8 = 0$$

$$P_1: s = -\frac{2}{2} \pm \sqrt{1-8}$$

$$s = -1 \pm \sqrt{7}j$$

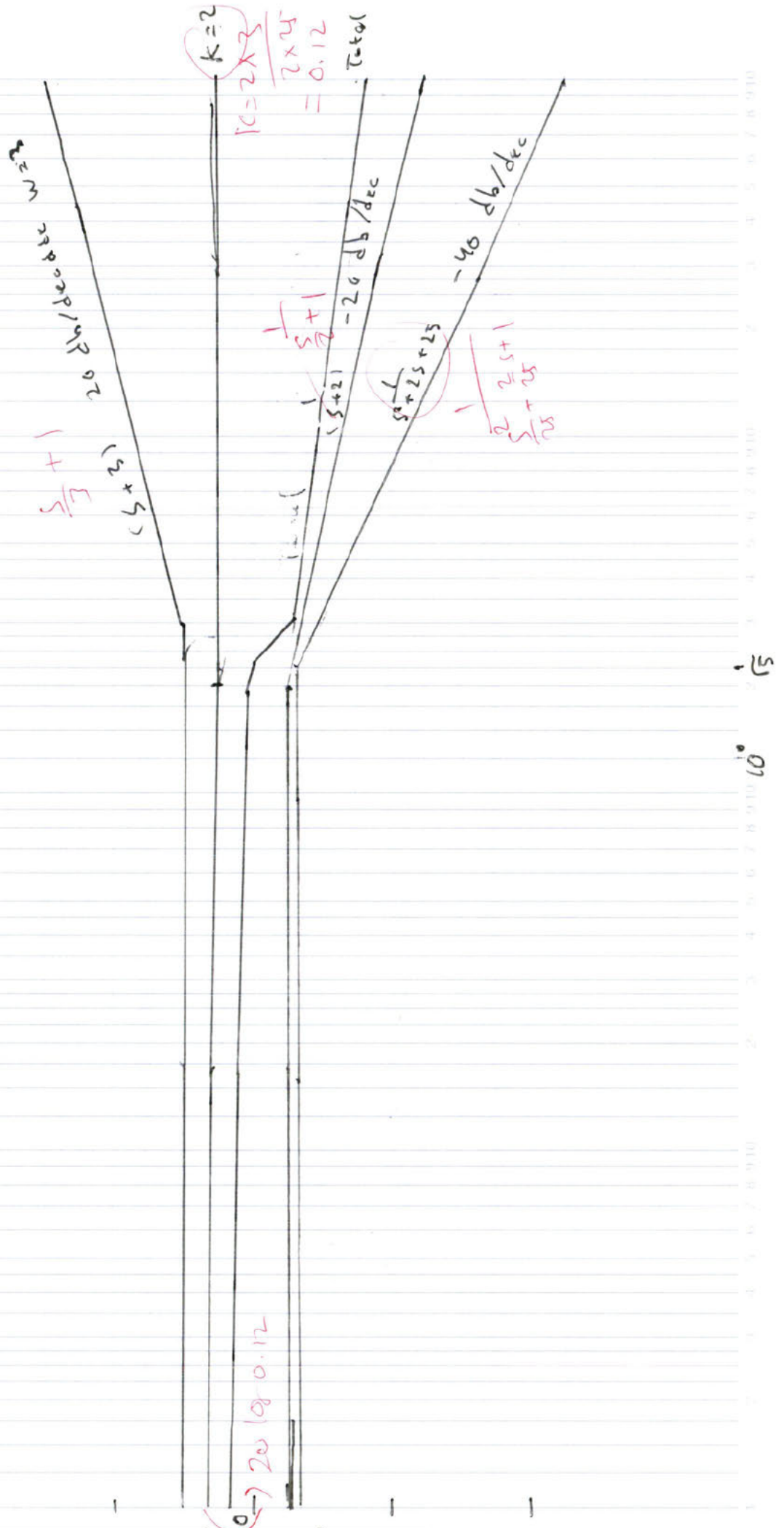
$$P = 0 \quad \checkmark$$

$$N = 2 \quad \checkmark$$

$$Z = 2 + 0 = 2 \quad \checkmark \quad Z > 0$$

unstable system.

5.11



Phase Plot

5.10

PERSONNUMBER:

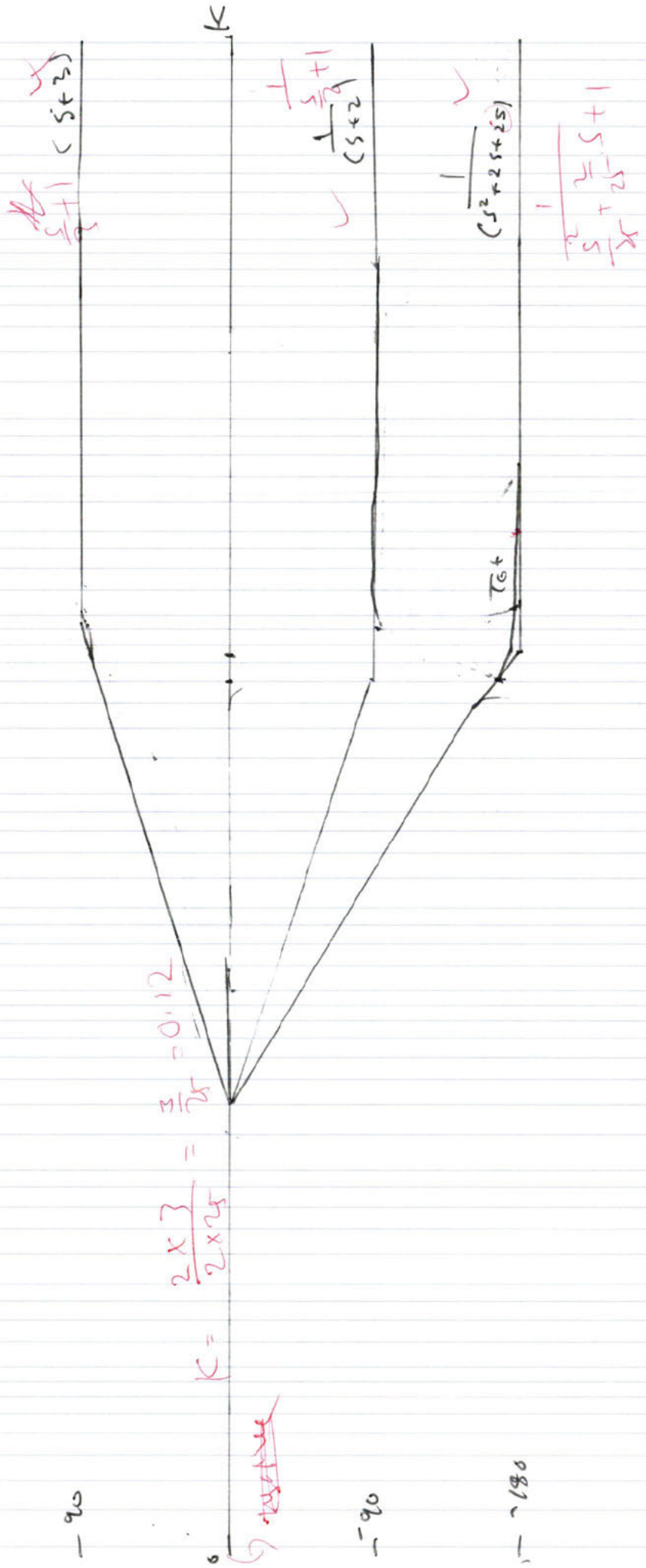
-180

-90

0

90

180



a) in the sketch paper.

b)

Phase cross over freq

No, ω_{gc} , never crossed -180

the $|G(\omega_{gc})| = \infty$

margin cross over freq:

crossed 0db roughly at $\sqrt{5}$

$\angle G(\sqrt{5}) \approx -175^\circ$

$-175 + 180 = 5^\circ$

*but your answer correct
based on your plot
but because your plot is wrong!*

c) if GM and PM are positive
then the system is stable.

Answer: system is stable.

Uppgift nr:

6

Poäng:

$0.8 + 2.5$

Lärarens

anteckning:

✓

0.3

✓

0.5

Uppgift nr:

7.

Poäng:

1.8

Lärarens
anteckning:

a)



$$T(s) = \frac{G(s)(k_p + k_d s)}{1 + G(s)(k_p + k_d s)}$$

$$G(s) = \frac{6}{s^2 + 2s + 6}$$

$$\begin{aligned} T(s) &= \frac{6(k_p + k_d s)}{s^2 + 2s + 6 + 6(k_p + k_d s)} \\ &= \frac{6(k_p + k_d s)}{s^2 + 2s + 6 + 6k_p + 6k_d s} \end{aligned}$$

$$T(s) = \frac{6(k_p + k_d s)}{s^2 + (2 + 6k_d)s + 6 + 6k_p}$$

Char equation = ?

$$2\omega_n^2 = 2 + 6k_d$$

$$\frac{(2\omega_n^2 - 2)}{6} = k_d = 2.666...7$$

$$6 + 6k_p = \omega_n^2$$

$$\frac{(\omega_n^2 - 6)}{6} = k_p = \frac{4}{6} = 0.666...7$$

b) it increases the damping ratio.

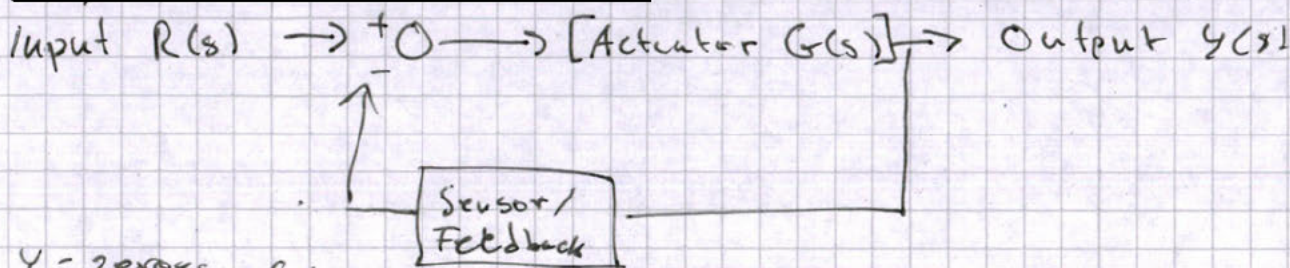
b?

0.8

0.5

0.5

φ



$$\frac{Y}{U} = \text{zeros} = \text{Poles} = G(s)$$

$$T(s) = \frac{G(s)H(s)}{1 + G(s)H(s)}$$

Stability: transfer funct.
 Poles in RHP = unstable
 LHP = stable
 State Space
 $\det(sI - A) = 0$

$20 \log(K)$, Constant gain factor.

$0 < \zeta < 1$ under damped

$\zeta > 1$ over damped

Phase cross over -180° $|G(j\omega_{pc})| = GM$

Gain cross over 0db $\angle(G(j\omega_{gc})) = PM$

$Z = N + P$ $P = \text{in RHP}$ $N = \text{Encirclement } (1,0)$

$Z > 0$ unstable if $GM \geq PM > 0$ stable sys.

Settle time $t_s = \frac{4}{\zeta \omega_n}$ (2% criterion)

$Z = 0$ stable

$\frac{3}{\zeta \omega_n}$ (5% criterion)

OverShoot $M_p = e^{\frac{-2\pi}{\sqrt{1-\zeta^2}}}$

Extra zero: LHP $\uparrow$ M_p

Extra zero: RHP $\downarrow$ M_p

Extra Pole: LHP $\uparrow$ t_p

t_r Rise time 10% $\rightarrow$ 90% of its final val

$$t_r \approx \frac{1.8}{\omega_n}$$

$$t_p \text{ Peak time} = \frac{\pi}{\omega_d} \quad \omega_d = \omega_n \sqrt{1-\zeta^2}$$

$$K_p = \lim_{s \rightarrow 0} T(s) \quad E_{ss} = \frac{1}{1+K_p} \quad (\text{closed systems})$$

$E_{ss} R(s)$ - output open systems
 for input step